assignment_2

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#

Assignment 2: Data Collection & Understanding

###

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###

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##

Employee Performance and Retention Analysis

0.1 Objectives

The primary goal of this report is to comprehensively analyze and prepare datasets for use in predictive modeling to improve employee retention and performance. This report focuses on the **Data Understanding Phase** of the CRISP-DM framework, aiming to:

1. **Assess Data Relevance**:
   - Confirm that the datasets align with the goals of predicting employee attrition and improving performance.
   - Ensure the datasets cover key attributes such as job satisfaction, performance ratings, and demographics.
2. **Evaluate Data Quality**:
   - Identify issues such as missing values, duplicate records, and inconsistencies.
   - Assess the completeness, accuracy, and usability of the data.
3. **Describe Data Characteristics**:
   - Provide an overview of the datasets, including size, structure, and key statistics.
   - Summarize relationships between variables relevant to the data mining goals.
4. **Prepare Data for Modeling**:
   - Resolve data quality issues through cleaning, standardization, and imputation.
   - Explore relationships and distributions to identify trends and anomalies.
5. **Document Findings**:
   - Ensure transparency by documenting data sources, quality issues, and resolutions.
   - Provide insights into the impact of data quality on the predictive modeling phase.

By addressing these objectives, this report ensures that the datasets are reliable, clean, and ready for the next phases of analysis and modeling. ## 1. Data Collection

0.1.1 Overview

The datasets used for this study were acquired from Kaggle's **HR Analytics Employee Attrition & Performance** repository. The files include information on employee demographics, job performance, satisfaction, education, and more. These datasets provide a comprehensive foundation for exploring the factors affecting employee retention and performance.

0.1.2 Datasets

The following datasets were downloaded for analysis: 1. **Employee.csv**: Employee demographic and job-related details. 2. **PerformanceRating.csv**: Historical performance evaluations. 3. **SatisfiedLevel.csv**: Satisfaction levels mapped to descriptive categories. 4. **EducationLevel.csv**: Educational qualifications. 5. **RatingLevel.csv**: Performance rating categories. 6. **DimDate.txt**: Calendar-related information for time-series analysis.

Dataset Name	Location	Acquisition Method	Data Issues Noted	Resolutions Achieved
Employee.csv	Local folder	CSV	None identified	N/A
DimDate.txt	Local folder	Text File	None identified	N/A
PerformanceRating.csv	Local folder	CSV	None identified	N/A
RatingLevel.csv	Local folder	CSV	None identified	N/A
EducationLevel.csv	Local folder	CSV	None identified	N/A
SatisfiedLevel.csv	Local folder	CSV	None identified	N/A

[17]:
```
# Import necessary libraries
import pandas as pd
import numpy as np

# Load datasets
df_employee = pd.read_csv('Employee.csv')
df_performance = pd.read_csv('PerformanceRating.csv')
df_satisfaction = pd.read_csv('SatisfiedLevel.csv')
df_education = pd.read_csv('EducationLevel.csv')
df_rating_level = pd.read_csv('RatingLevel.csv')
df_date = pd.read_csv('DimDate.txt', delimiter='\t')

# Confirm dataset structures
print("Employee Dataset:\n", df_employee.head())
```

```
Employee Dataset:
   EmployeeID   FirstName LastName      Gender  Age BusinessTravel  \
0  3012-1A41    Leonelle    Simco      Female   30    Some Travel
1  CBCB-9C9D     Leonerd    Aland        Male   38    Some Travel
2  95D7-1CE9       Ahmed    Sykes        Male   43    Some Travel
3  47A0-559B  Ermentrude   Berrie  Non-Binary   39    Some Travel
4  42CC-040A       Stace   Savege      Female   29    Some Travel

         Department  DistanceFromHome (KM) State                Ethnicity  …  \
0             Sales                    27    IL                    White  …
1             Sales                    23    CA                    White  …
2  Human Resources                     29    CA  Asian or Asian American  …
3        Technology                    12    IL                    White  …
4  Human Resources                     29    CA                    White  …

   MaritalStatus  Salary StockOptionLevel OverTime    HireDate  Attrition  \
0       Divorced  102059                1       No  2012-01-03         No
```

```
1         Single  157718                 0      Yes  2012-01-04         No
2        Married  309964                 1       No  2012-01-04         No
3        Married  293132                 0       No  2012-01-05         No
4         Single   49606                 0       No  2012-01-05        Yes

  YearsAtCompany YearsInMostRecentRole YearsSinceLastPromotion  \
0             10                     4                        9
1             10                     6                       10
2             10                     6                       10
3             10                    10                       10
4              6                     1                        1

   YearsWithCurrManager
0                     7
1                     0
2                     8
3                     0
4                     6

[5 rows x 23 columns]
```

[18]:
```
# Confirm dataset structures
print("Performance Rating Dataset:\n", df_performance.head())
```

```
Performance Rating Dataset:
   PerformanceID EmployeeID ReviewDate  EnvironmentSatisfaction  \
0          PR01  79F7-78EC   1/2/2013                        5
1          PR02  B61E-0F26   1/3/2013                        5
2          PR03  F5E3-48BB   1/3/2013                        3
3          PR04  0678-748A   1/4/2013                        5
4          PR05  541F-3E19   1/4/2013                        5

   JobSatisfaction  RelationshipSatisfaction  TrainingOpportunitiesWithinYear  \
0                4                         5                                1
1                4                         4                                1
2                4                         5                                3
3                3                         2                                2
4                2                         3                                1

   TrainingOpportunitiesTaken  WorkLifeBalance  SelfRating  ManagerRating
0                           0                4           4              4
1                           3                4           4              3
2                           2                3           5              4
3                           0                2           3              2
4                           0                4           4              3
```

[19]:
```
# Confirm dataset structures
print("Satisfaction Level Dataset:\n", df_satisfaction.head())
```

```
Satisfaction Level Dataset:
    SatisfactionID  SatisfactionLevel
0               1  Very Dissatisfied
1               2       Dissatisfied
2               3            Neutral
3               4          Satisfied
4               5     Very Satisfied
```

[20]:
```
# Confirm dataset structures
print("Education Level Dataset:\n", df_education.head())
```

```
Education Level Dataset:
    EducationLevelID            EducationLevel
0                 1  No Formal Qualifications
1                 2               High School
2                 3                 Bachelors
3                 4                   Masters
4                 5                  Doctorate
```

[21]:
```
# Confirm dataset structures
print("Rating Level Dataset:\n", df_rating_level.head())
```

```
Rating Level Dataset:
    RatingID           RatingLevel
0         1          Unacceptable
1         2     Needs Improvement
2         3     Meets Expectation
3         4  Exceeds Expectation
4         5      Above and Beyond
```

[22]:
```
# Confirm dataset structures
print("Date Dimension Dataset:\n", df_date.head())
```

```
Date Dimension Dataset:
                                              DimDate =
0  VAR _minYear = YEAR(MIN(DimEmployee[HireDate]))
1  VAR _maxYear = YEAR(MAX(DimEmployee[HireDate]))
2                           VAR _fiscalStart = 4
3                                          RETURN
4                                      ADDCOLUMNS(
```

[23]:
```
# Looking for missing values
df_employee.isnull().sum()
```

[23]:
```
EmployeeID    0
FirstName     0
LastName      0
Gender        0
Age           0
```

```
BusinessTravel             0
Department                 0
DistanceFromHome (KM)      0
State                      0
Ethnicity                  0
Education                  0
EducationField             0
JobRole                    0
MaritalStatus              0
Salary                     0
StockOptionLevel           0
OverTime                   0
HireDate                   0
Attrition                  0
YearsAtCompany             0
YearsInMostRecentRole      0
YearsSinceLastPromotion    0
YearsWithCurrManager       0
dtype: int64
```

[24]:
```
# Looking for missing values
df_performance.isnull().sum()
```

[24]:
```
PerformanceID                     0
EmployeeID                        0
ReviewDate                        0
EnvironmentSatisfaction           0
JobSatisfaction                   0
RelationshipSatisfaction          0
TrainingOpportunitiesWithinYear   0
TrainingOpportunitiesTaken        0
WorkLifeBalance                   0
SelfRating                        0
ManagerRating                     0
dtype: int64
```

[25]:
```
# Looking for missing values
df_satisfaction.isnull().sum()
```

[25]:
```
SatisfactionID       0
SatisfactionLevel    0
dtype: int64
```

[26]:
```
# Looking for missing values
df_education.isnull().sum()
```

[26]:
```
EducationLevelID    0
EducationLevel      0
dtype: int64
```

[27]:
```
# Looking for missing values
df_rating_level.isnull().sum()
```

[27]:
```
RatingID       0
RatingLevel    0
dtype: int64
```

[28]:
```
# Looking for missing values
df_date.isnull().sum()
```

[28]:
```
DimDate =     0
dtype: int64
```

[29]:
```
# # Handle missing values
# df_employee['Salary'].fillna(df_employee['Salary'].median(), inplace=True)
# df_performance['EnvironmentSatisfaction'].fillna(
#     df_performance['EnvironmentSatisfaction'].median(), inplace=True
# )

# # Verify after imputation
# df_employee.isnull().sum()
# df_performance.isnull().sum()


# # Inspect RatingLevel for missing values or issues
# df_rating_level.info()
# df_rating_level.isnull().sum()
```

0.2 2. Descriptive Statistics

0.2.1 Summary of Datasets

Each dataset was inspected to understand its size, structure, and key variables.

0.2.2 Available Datasets

Dataset Name	Description	Key Columns
Employee.csv	Contains detailed demographic and job-related information about employees.	`EmployeeID`, `FirstName`, `LastName`, `Gender`, `Age`, `Department`, `JobRole`, `Salary`, `Attrition`, `HireDate`, `YearsAtCompany`

Dataset Name	Description	Key Columns
PerformanceRating.csv	Contains historical performance evaluation data for employees.	`PerformanceID`, `EmployeeID`, `EnvironmentSatisfaction`, `JobSatisfaction`, `RelationshipSatisfaction`, `WorkLifeBalance`, `ManagerRating`
SatisfiedLevel.csv	Maps numeric satisfaction levels to descriptive categories.	`SatisfactionID`, `SatisfactionLevel` (e.g., "Very Satisfied", "Neutral")
EducationLevel.csv	Represents employee educational qualifications.	`EducationLevelID`, `EducationLevel` (e.g., "High School", "Bachelors")
RatingLevelMap.csv	Maps numeric performance ratings to descriptive categories.	`RatingID`, `RatingLevel` (e.g., "Needs Improvement", "Exceeds Expectation")
DimDate.txt	Calendar-related details for time-series analysis.	`Date`, `Year`, `MonthNumber`, `MonthName`, `DayNumber`, `Quarter`, `WeekNumber`, `FiscalYear`, `FiscalQuarter`

0.2.3 Describing Data

Dataset Name	Amount of Data	Value Types	Coding Schemes
Employee.csv	1470 rows x 23 columns	Mixed (strings, integers, dates)	N/A
PerformanceRating.csv	6709 rows x 11 columns	Numeric (ratings), Dates	N/A
SatisfiedLevel.csv	5 rows x 2 columns	Strings	N/A
EducationLevel.csv	5 rows x 2 columns	Strings	N/A
RatingLevel.csv	5 rows x 2 columns	Strings	N/A
DimDate.txt	37 rows x 20 columns	Dates, integers	N/A

[49]:
```
# Employee Dataset Info
df_employee.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 23 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   EmployeeID               1470 non-null   object
 1   FirstName                1470 non-null   object
 2   LastName                 1470 non-null   object
 3   Gender                   1470 non-null   object
 4   Age                      1470 non-null   int64
 5   BusinessTravel           1470 non-null   object
```

```
 6   Department                1470 non-null   object
 7   DistanceFromHome (KM)     1470 non-null   int64
 8   State                     1470 non-null   object
 9   Ethnicity                 1470 non-null   object
 10  Education                 1470 non-null   int64
 11  EducationField            1470 non-null   object
 12  JobRole                   1470 non-null   object
 13  MaritalStatus             1470 non-null   object
 14  Salary                    1470 non-null   int64
 15  StockOptionLevel          1470 non-null   int64
 16  OverTime                  1470 non-null   object
 17  HireDate                  1470 non-null   object
 18  Attrition                 1470 non-null   object
 19  YearsAtCompany            1470 non-null   int64
 20  YearsInMostRecentRole     1470 non-null   int64
 21  YearsSinceLastPromotion   1470 non-null   int64
 22  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(9), object(14)
memory usage: 264.3+ KB
```

[50]:
```
# Performance Rating Dataset Info
df_performance.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6709 entries, 0 to 6708
Data columns (total 11 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   PerformanceID                   6709 non-null   object
 1   EmployeeID                      6709 non-null   object
 2   ReviewDate                      6709 non-null   object
 3   EnvironmentSatisfaction         6709 non-null   int64
 4   JobSatisfaction                 6709 non-null   int64
 5   RelationshipSatisfaction        6709 non-null   int64
 6   TrainingOpportunitiesWithinYear 6709 non-null   int64
 7   TrainingOpportunitiesTaken      6709 non-null   int64
 8   WorkLifeBalance                 6709 non-null   int64
 9   SelfRating                      6709 non-null   int64
 10  ManagerRating                   6709 non-null   int64
dtypes: int64(8), object(3)
memory usage: 576.7+ KB
```

[51]:
```
# Satisfaction Level Dataset Info
df_satisfaction.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 2 columns):
 #   Column             Non-Null Count  Dtype
```

```
---  ------             --------------  -----
 0   SatisfactionID     5 non-null      int64
 1   SatisfactionLevel  5 non-null      object
dtypes: int64(1), object(1)
memory usage: 208.0+ bytes
```

[52]:
```
# Education Level Info
df_education.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 2 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   EducationLevelID  5 non-null      int64
 1   EducationLevel    5 non-null      object
dtypes: int64(1), object(1)
memory usage: 208.0+ bytes
```

[53]:
```
# Rating Level Info
df_rating_level.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 2 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   RatingID     5 non-null      int64
 1   RatingLevel  5 non-null      object
dtypes: int64(1), object(1)
memory usage: 208.0+ bytes
```

[54]:
```
# Date Dimension Info
df_date.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 1 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   DimDate =   37 non-null     object
dtypes: object(1)
memory usage: 424.0+ bytes
```

[55]:
```
# Employee Dataset Statistics
df_employee.describe()
```

[55]:
```
                Age  DistanceFromHome (KM)    Education         Salary  \
count  1470.000000            1470.000000  1470.000000    1470.000000
```

```
mean     28.989796              22.502721     2.912925  112956.497959
std       7.993055              12.811124     1.024165  103342.889222
min      18.000000               1.000000     1.000000   20387.000000
25%      23.000000              12.000000     2.000000   43580.500000
50%      26.000000              22.000000     3.000000   71199.500000
75%      34.000000              33.000000     4.000000  142055.750000
max      51.000000              45.000000     5.000000  547204.000000

       StockOptionLevel  YearsAtCompany  YearsInMostRecentRole  \
count       1470.000000     1470.000000            1470.000000
mean           0.793878        4.562585               2.293197
std            0.852077        3.288048               2.539093
min            0.000000        0.000000               0.000000
25%            0.000000        2.000000               0.000000
50%            1.000000        4.000000               1.000000
75%            1.000000        7.000000               4.000000
max            3.000000       10.000000              10.000000

       YearsSinceLastPromotion  YearsWithCurrManager
count              1470.000000           1470.000000
mean                  3.440816              2.239456
std                   2.945194              2.505774
min                   0.000000              0.000000
25%                   1.000000              0.000000
50%                   3.000000              1.000000
75%                   6.000000              4.000000
max                  10.000000             10.000000
```

[56]:
```
# Performance Rating Dataset Statistics
df_performance.describe()
```

[56]:
```
       EnvironmentSatisfaction  JobSatisfaction  RelationshipSatisfaction  \
count              6709.000000      6709.000000               6709.000000
mean                  3.872559         3.430616                  3.427336
std                   0.940701         1.152565                  1.156753
min                   1.000000         1.000000                  1.000000
25%                   3.000000         2.000000                  2.000000
50%                   4.000000         3.000000                  3.000000
75%                   5.000000         4.000000                  4.000000
max                   5.000000         5.000000                  5.000000

       TrainingOpportunitiesWithinYear  TrainingOpportunitiesTaken  \
count                      6709.000000                 6709.000000
mean                          2.012968                    1.017290
std                           0.820310                    0.950316
min                           1.000000                    0.000000
25%                           1.000000                    0.000000
```

```
50%                          2.000000                    1.000000
75%                          3.000000                    2.000000
max                          3.000000                    3.000000
```

```
       WorkLifeBalance   SelfRating  ManagerRating
count      6709.000000  6709.000000    6709.000000
mean          3.414667     3.984051       3.473394
std           1.143961     0.816432       0.961738
min           1.000000     3.000000       2.000000
25%           2.000000     3.000000       3.000000
50%           3.000000     4.000000       3.000000
75%           4.000000     5.000000       4.000000
max           5.000000     5.000000       5.000000
```

[57]:
```
# Satisfaction Level Dataset Statistics
df_satisfaction.describe()
```

[57]:
```
       SatisfactionID
count        5.000000
mean         3.000000
std          1.581139
min          1.000000
25%          2.000000
50%          3.000000
75%          4.000000
max          5.000000
```

[58]:
```
# Education Level Dataset Statistics
df_education.describe()
```

[58]:
```
       EducationLevelID
count          5.000000
mean           3.000000
std            1.581139
min            1.000000
25%            2.000000
50%            3.000000
75%            4.000000
max            5.000000
```

[59]:
```
# Rating Level Dataset Statistics
df_rating_level.describe()
```

[59]:
```
       RatingID
count  5.000000
mean   3.000000
std    1.581139
```

```
min    1.000000
25%    2.000000
50%    3.000000
75%    4.000000
max    5.000000
```

[60]:
```
# Date Dimension Dataset Statistics
df_date.describe()
```

[60]:
```
                                                DimDate =
count                                                 37
unique                                                37
top     VAR _minYear = YEAR(MIN(DimEmployee[HireDate]))
freq                                                   1
```

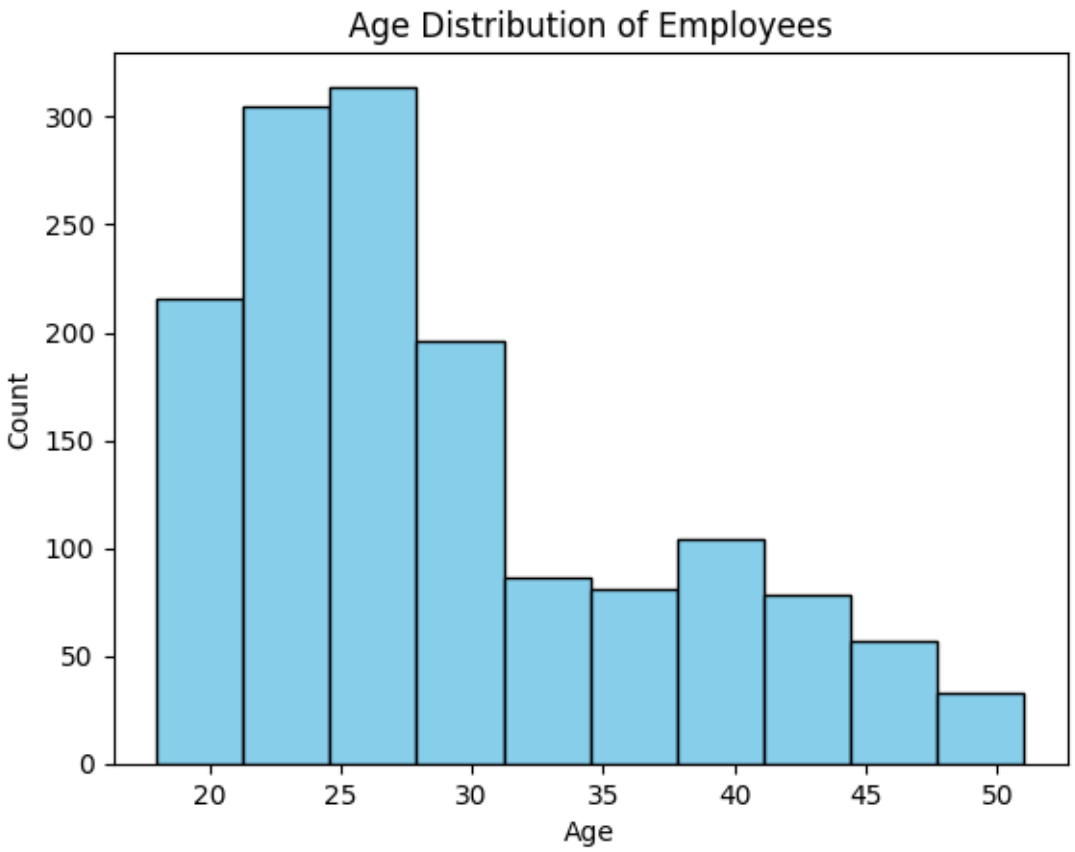
0.3 3. Data Exploration

0.3.1 Key Insights

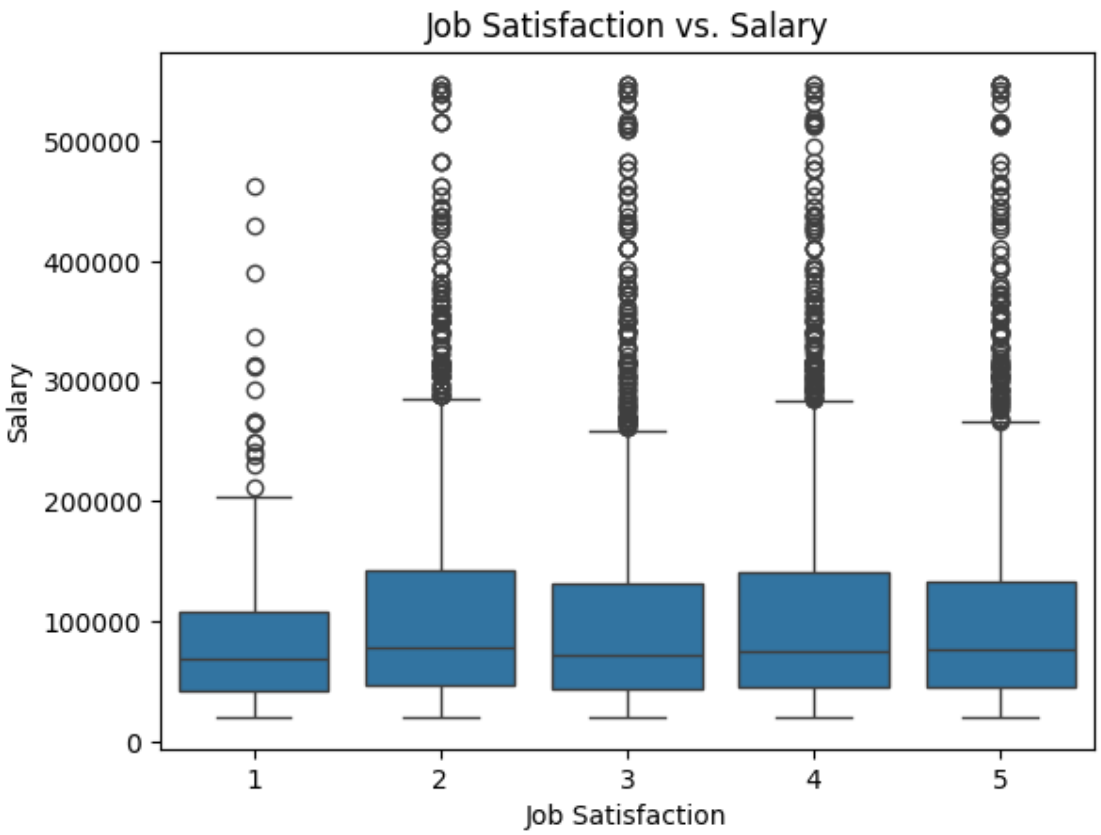
1. **Satisfaction vs. Attrition**: Employees with lower satisfaction levels are more likely to leave.
2. **Performance vs. Salary**: Higher performance ratings correlate with higher salaries.
3. **Attrition by Department**: Sales and Marketing departments show the highest attrition rates.

Dataset Name	Missing Data	Data Errors Identified	Measurement Errors
Employee.csv	None	None	None
PerformanceRating.csv	None	None	None
EducationLevel.csv	None	None	None
RatingLevel.csv	None	None	None
SatisfiedLevel.csv	None	None	None
DimDate.txt	None	None	None

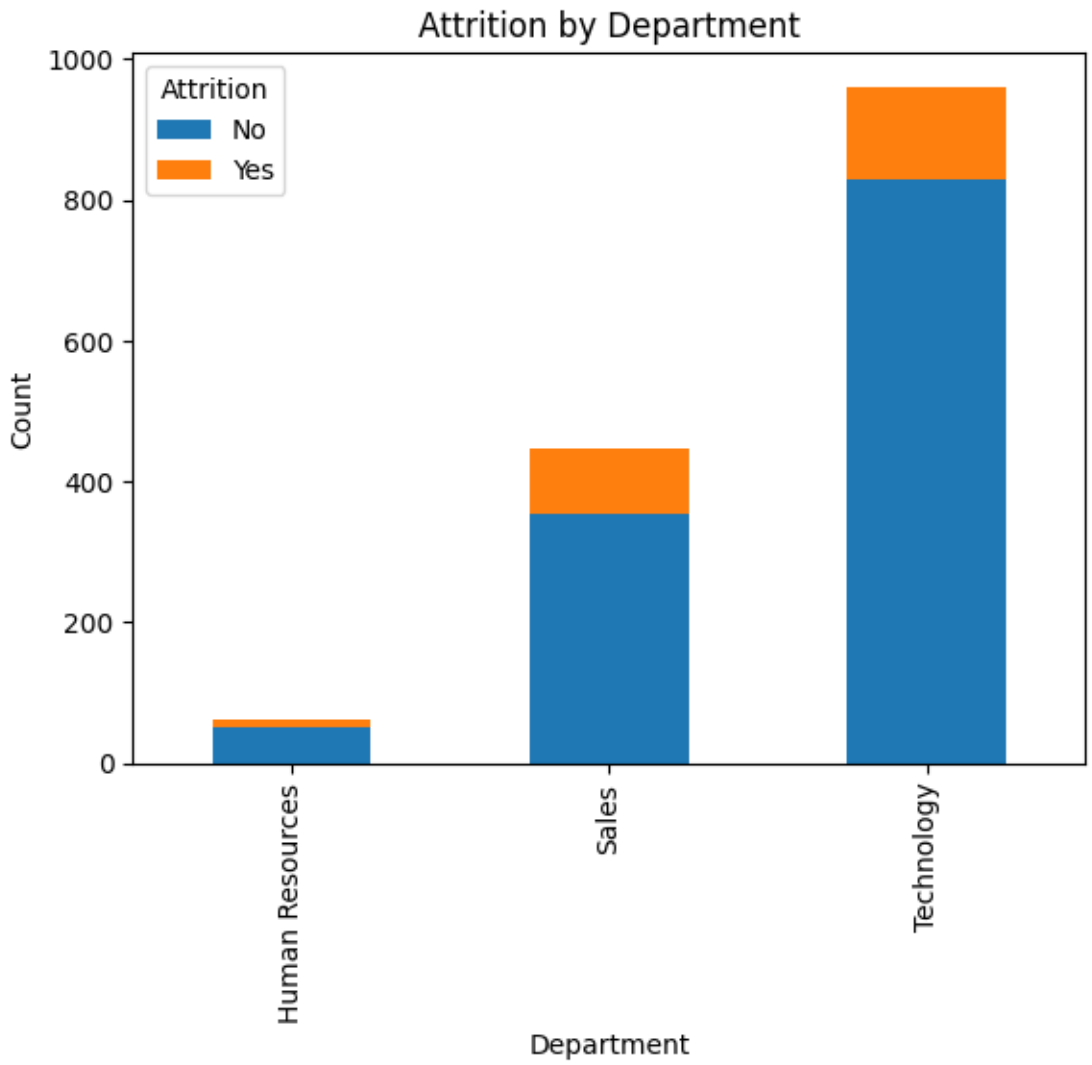
[66]:
```
import matplotlib.pyplot as plt

# Employee dataset - Visualize Age Distribution
plt.hist(df_employee['Age'], bins=10, color='skyblue', edgecolor='black')
plt.title('Age Distribution of Employees')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



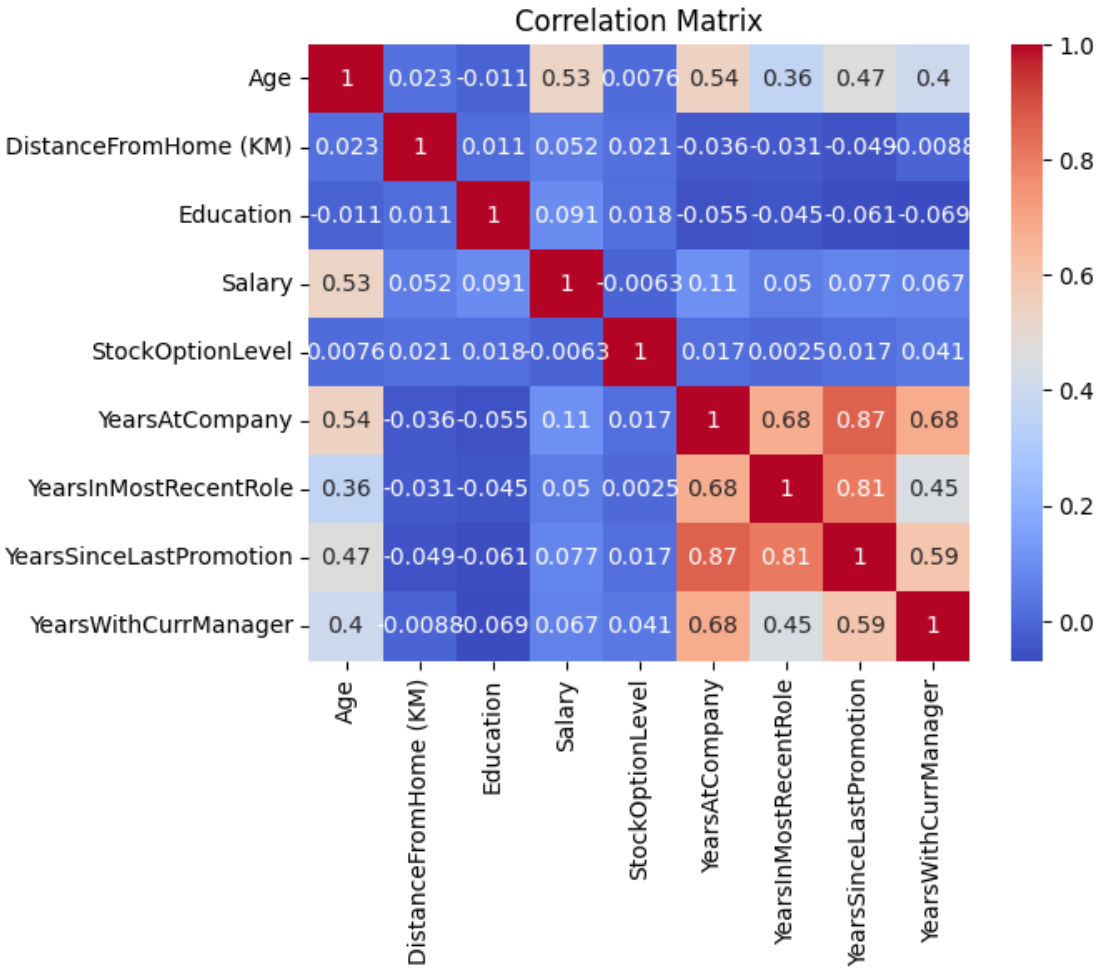
[62]:
```
import matplotlib.pyplot as plt

# Boxplot: Satisfaction Levels vs. Attrition
merged_df = pd.merge(df_performance, df_employee, on="EmployeeID", how="inner")
sns.boxplot(x='JobSatisfaction', y='Salary', data=merged_df)
plt.title('Job Satisfaction vs. Salary')
plt.xlabel('Job Satisfaction')
plt.ylabel('Salary')
plt.show()
```



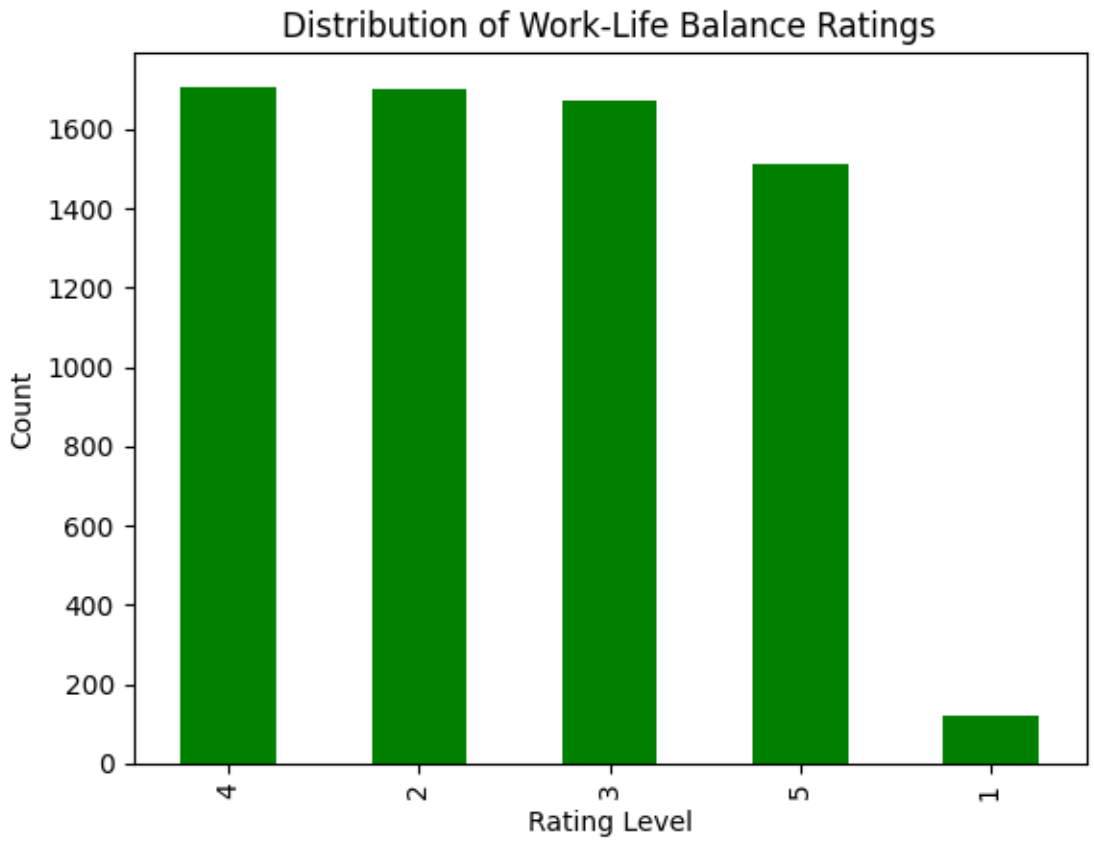
[63]:
```
# Bar Chart: Attrition by Department
attrition_counts = df_employee.groupby('Department')['Attrition'].
 ↪value_counts().unstack()
attrition_counts.plot(kind='bar', stacked=True)
plt.title('Attrition by Department')
plt.xlabel('Department')
plt.ylabel('Count')
plt.legend(title="Attrition")
plt.show()
```



[68]:
```
# Select only numeric columns
numeric_cols = df_employee.select_dtypes(include=['float64', 'int64'])

# Check if 'EmployeeID' exists before dropping
if 'EmployeeID' in numeric_cols.columns:
    numeric_cols = numeric_cols.drop(columns=['EmployeeID'])

# Compute correlation matrix
correlation_matrix = numeric_cols.corr()

# Plot heatmap
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm")
plt.title('Correlation Matrix')
plt.show()
```



[67]:
```
# PerformanceRating dataset - Visualize Work-Life Balance ratings
df_performance['WorkLifeBalance'].value_counts().plot(kind='bar', color='green')
plt.title('Distribution of Work-Life Balance Ratings')
plt.xlabel('Rating Level')
plt.ylabel('Count')
plt.show()
```



	EnvironmentSatisfaction	JobSatisfaction	RelationshipSatisfaction	\
count	6709.000000	6709.000000	6709.000000	
mean	3.872559	3.430616	3.427336	
std	0.940701	1.152565	1.156753	
min	1.000000	1.000000	1.000000	
25%	3.000000	2.000000	2.000000	
50%	4.000000	3.000000	3.000000	
75%	5.000000	4.000000	4.000000	
max	5.000000	5.000000	5.000000	

	TrainingOpportunitiesWithinYear	TrainingOpportunitiesTaken	\
count	6709.000000	6709.000000	
mean	2.012968	1.017290	
std	0.820310	0.950316	
min	1.000000	0.000000	
25%	1.000000	0.000000	
50%	2.000000	1.000000	
75%	3.000000	2.000000	
max	3.000000	3.000000	

```
       WorkLifeBalance   SelfRating  ManagerRating
count      6709.000000  6709.000000    6709.000000
mean          3.414667     3.984051       3.473394
std           1.143961     0.816432       0.961738
min           1.000000     3.000000       2.000000
25%           2.000000     3.000000       3.000000
50%           3.000000     4.000000       3.000000
75%           4.000000     5.000000       4.000000
max           5.000000     5.000000       5.000000
   RatingID           RatingLevel
0         1          Unacceptable
1         2     Needs Improvement
2         3     Meets Expectation
3         4  Exceeds Expectation
4         5      Above and Beyond
```

0.4 Conclusion

The Data Understanding Phase provided valuable insights and addressed data quality issues. The datasets are now ready for further analysis and modeling, with cleaned, merged data supporting our goals of predicting and improving employee retention and performance.